NATHAN SUBRAHMANIAN

nathansubrahmanian@gmail.com | 301-525-2704 | https://www.linkedin.com/in/nathan-subrahmanian-6b9a90201 | **US Citizen**

EDUCATION

Brandeis University (GPA: 3.6/4; GRE (340): 170 Quant, 162 Verbal) — **Waltham, MA**

M.S. in Computational Linguistics — August 2025 - May 2026

Bachelor of Science in Computer Science; Bachelors of Arts in Business — August 2021 - May 2025

- **Relevant Coursework:** Statistical ML, Practical ML, Computer Vision, NLP (~6 courses), Unsupervised Learning, Embedded Systems, Software Engineering, ASR, Operating Systems, Sports Analytics, Math for ML, Info Extraction

APPLIED AI AND MACHINE LEARNING EXPERIENCE

Helios Healthcare Advisors – Lead ML Intern — June 2026 – Present

- Created a Python/PostgreSQL ML pipeline forecasting quality-of-care deltas across 5k+ healthcare facilities post-acquisition
- Architected a feature pipeline utilizing similarity-weighted portfolio embeddings and normalization to reduce temporal drift

Northwestern University – AI/ML Graduate Research Assistant, NSAIL Lab — May 2025 – Present

- Built and evaluated machine learning systems for detecting covert social influence campaigns, including RL-based adaptive social bots and human-AI hybrid detection workflows
- Implemented bot detection models and evaluation pipelines to compare human, model, and ensemble performance across ~13 experimental conditions
- Analyzed demographic and subgroup-level performance differences using statistical testing to identify bias, robustness, and failure modes in AI-assisted decision systems
- Worked with RL-based bot behavior simulations to study how adaptive adversaries evade detection systems
- Translated experimental research questions into reproducible ML workflows involving preprocessing, inference, evaluation, and statistical validation
- **Human, AI, and Hybrid Ensembles for Detection of Adaptive, RL-based Social Bots (accepted ACM TWEB'26)**

Georgia Institute of Technology – AI/ML Undergraduate Researcher, CLAWS Lab — May 2022 – January 2025

- Fine-tuned BERT-based transformer models for misinformation + social-correction classification on large Twitter datasets
- Built NLP pipelines for preprocessing, labeling, feature extraction, training, and evaluation across millions of tweets and users
- Developed classification workflows to identify counter-misinformation behavior, user response patterns in social media conversations
- Performed large-scale behavioral analysis w/ Python, pandas, NumPy, and statistical tests to evaluate demographic differences in online social correction behavior
- **Characterizing and Predicting Social Correction on Twitter (published in ACM's WebSci'23)**

AI ENGINEERING PROJECTS

Congressional Bill Constraint-based Prediction and Interactive ML Dashboard

- Built a data pipeline around the Congress.gov API to collect House members, roll-call votes, and bill text + metadata
- Parsed XML bill text into clean, modeling-ready documents for downstream NLP and classification tasks
- Developed predictive modeling workflows to estimate legislator voting behavior, combined with a constraint-based layer to identify likely voting coalitions, disagreement points, and policy/political constraints
- Designed an interactive dashboard for exploring predictions and what-if scenarios for improved model interpretability

LEADERSHIP EXPERIENCE

Brandeis University — **Waltham, MA**

Teaching Assistant: Introduction to Data Analytics with Excel — August 2023 – December 2023

- Assisted Professor Gizem Nemutlu in ensuring 40+ students understood statistics during lecture and grading assignments
- Held office hours to help students prepare for quizzes, learn material, and develop study skills

Technical / ML Skills: Python, NumPy/pandas, Java, SKlearn, PyTorch, Tensorflow, HuggingFace, PostgreSQL, HTML/CSS, Excel, Deep Learning, traditional ML, CNNs, LLMs and NLP tasks, RLHF